



Four Sigma

**Quantitative Finance
Hackathon
Team 1**

May 2, 2026



Meet the Team

Shikha Sharma



MIDS Graduate, May 2026
Tech Lead, UHC (past)

Goal : Gain hands on experience building AI/ML applications while collaborating with others.

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Justin Sterling



MIDS Graduate, May 2026
Financial Engineer, CFA FRM

Goal - Get more hands-on experience with end-to-end data science projects, learn more about intraday trading strategy

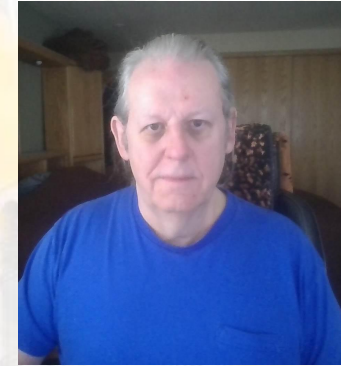
Jennifer Nishimura



2nd Semester, Graduating FA2026

Goal: To collaborate with others and gain more hands-on project experience.

Daniel Davies



MIDS Alumni Summer 2017
Electrical Engineer, Retired

Goa: Learn more about FinTech and collaborate with others

Feature Engineering

Per Ticker 30min Features

8 Features

- Momentum
- Volume Change
- Bar Range
- Rolling mean 5
- Rolling STD 5
- Rolling Volume Ratio
- Rolling Mean 13
- Rolling STD 13

Cross Sectional Features

7 Features

- Momentum Z
- Momentum rank
- Bar Range Z
- Volume Change Z
- Rolling Mean Z
- Rolling Mean 13 Z
- Rolling Volume Ratio Z

Sector Relative Features

2 Features

- Momentum relative to sector
- Momentum Sector Z

5 Min Microstructure

4 Features

- Intraday STD
- Close Position
- Volume
- Concentration
- Tail Strength

Extras

5 Features

- Minutes from Open
- Minutes to Close
- Dollar Volume
- VWAP
- Gap Return

Final Submission - Ensemble

Light GBM Regressor

- n_estimators - 1500
- learning_rate - 0.05
- num_leaves - 31
- min_child_samples - 500
- feature_fraction - 0.9
- bagging_fraction - 0.9
- bagging_freq - 1

XGBoost

- n_estimators - 1500
- learning_rate - 0.05
- max_depth - 6
- min_child_weight - 10
- subsample - 0.9
- colsample_bytree - 0.9
- tree_method - "hist"
- eval_metric - "rmse"

Random Forest

- n_estimators - 200
- max_depth - 10
- min_samples_leaf - 500
- max_features - sqrt
- sample_frac - 0.3

- Equal-weighted 3-tree ensemble.
- Originally, only kept LGB+RF, but with train_start=2015 (full 10y window), XGB earns its slot.
- Equal weights beats LGB+RF-only by +0.0003 mean IC and +0.002 IR.
- ENet was attempted but remained too weak to include.

Final Submission - Results

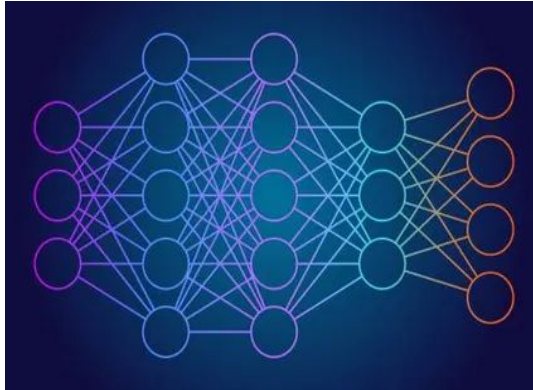
3-Tree Ensemble

- **IC Mean - .0727**
- IC Standard Dev - 0.1003
- IR - 0.724
- Pct Pos- 0.797

- Equal-weighted 3-tree ensemble (LGB + XGB + RF, 1/3 each) on 26 features with rank target.
- Trained on 2015+ data. Standalones: LGB +0.0721/IR 0.725, XGB +0.0723/IR 0.727, RF +0.0696/IR 0.683

Other Analysis

More Features

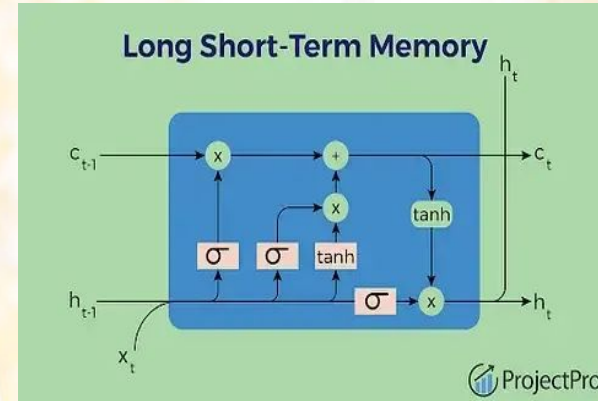


Features/Methods Not Used in Submission

- Compute-limits and time-constraints
- Technical indicators (50 day, 200 day, crossings)
- Candlestick “shadow ratios”
- Hierarchical/Other forms of clustering
- Exponential weighted moving averages

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LSTM



Training Incomplete

- Compute-limits and time-constraints
- Removed features
- Reduced number of layers, size
- Post-2010 dataset with 5-10 features used
- **Use subset of stocks based on clusters?**
- **Use only certain time periods?**

Lessons Learned

- Aligning the loss to the eval metric
- Intra bar micro features (5 min)
- Cross sectional normalization
- Polar library for large data



Conclusion

Takeaways:

- It was great collaborating with a team of people who hadn't worked together before.
- We learned a lot about finance-related features and the extensive feature engineering others have developed over time.
- Compute limitations were a major constraint given the size of the dataset.

**A BIG thank you to the Four Sigma Organizers for this opportunity!!
We learned a lot!**

Placeholder Background Slide

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